

GANTANG MASIMUJA

+6281221595529 | gmasimuja@gmail.com | https://www.linkedin.com/in/gantang-masimuja-b9a122173/ | https://portofoliogm.vercel.app/

Pancot Lor, RT01, RW06, Tawangmangu

Geotechnical & Geophysical Engineer with a Master's Degree from Institut Teknologi Bandung (ITB) and 5+ years of experience in subsurface investigations for energy, geothermal, and mining infrastructure. Demonstrated a strong track record of leading large-scale field operations with project budgets reaching IDR 1 Billion. Adept at merging civil/geophysical engineering expertise with modern technologies, including AI Agents, Machine Learning (Genetic Algorithms), and workflow automation (n8n) to accelerate data processing, increase feasibility study accuracy by up to 50%, and optimize operational efficiency. Ready to deliver data-driven and AI-powered solutions to global-scale engineering projects.

Work Experiences

Earth Physics Laboratory ITB - **Bandung, Indonesia** Oct 2019 - Present

Geophysical & Project Engineer

- Led and executed 15+ national-scale geophysical and geotechnical investigation projects for energy infrastructure, power transmission, and gas pipelines, achieving a 100% on-time project delivery rate.
- Developed and deployed geophysical data processing automation pipelines utilizing Python and n8n, successfully reducing field data extraction and reporting time by 40%.
- Conducted integrated feasibility studies for high-value renewable energy facilities, improving the accuracy of subsurface modeling for top-tier clients (e.g., Star Energy, PLN).
- Managed cross-functional teams of 10-15 field personnel and technicians, overseeing heavy equipment mobilization and ensuring strict HSE compliance with a 0 Lost Time Injury (Zero Accident) record.
- Designed AI-driven workflows to accelerate management decision-making through automated interpretation of raw geotechnical data into BOQ documents and technical reports.

HIGHLIGHTED ENGINEERING & AUTOMATION PROJECTS

- Geophysical Software Engineering: Designed and deployed a web application using Python and Django to automate the processing and visualization of Multichannel Analysis of Surface Waves (MASW) data.
- Seismic Downhole Feasibility Study (Gunung Salak & Darajat Geothermal Field): Analyzed hundreds of subsurface data points for Star Energy Geothermal, providing critical recommendations that saved potential drilling re-work costs by up to 15%.
- Tuban Gas Pipeline Subsurface Characterization: Executed Seismic Downhole & Resistivity surveys to design a 60 km offshore pipeline route, ensuring infrastructure safety from geological anomalies.
- Renewable Energy Infrastructure Modeling (Pulau Bulan Solar Power): Executed Geoelectrical & Thermal Resistivity investigations across a 100+ Hectare area to support optimal solar panel foundation design.
- Workflow Automation for EdTech & Operations: Built a fully integrated digital education platform ecosystem (using Next.js) featuring database and user registration automation, increasing operational administrative efficiency by 60%.

Education Level

Institut Teknologi Bandung - **Bandung, Indonesia** Aug 2023 - Aug 2025

Master of Geotechnical Engineering, 3.66/4.00

- Thesis: Development of MASW Data Inversion with Genetic Algorithm Method to Support Geotechnical Analysis
- Highlight: Implemented Machine Learning (Genetic Algorithms) and numerical computation for precise underground structure modeling, directly supporting geothermal infrastructure evaluations.

Institut Teknologi Bandung - **Bandung, Indonesia** Aug 2015 - Oct 2019

Bachelor of Physics, 3.08/4.00

Other Experiences

Phiwiki Team - **Bandung, Indonesia** Feb 2018 - Mar 2019

Head of Production Division

- Optimized supply chain operations and vendor negotiations using data-driven planning, successfully distributing >2,500 books per semester and generating a net profit of >IDR 35 Million (improving margin efficiency by 20%).

CORE SKILLS

- **AI, Automation & Software**: AI Agents Development, n8n Workflow Automation, Python, Django, Next.js, C++, Cloud AI Integrations (Azure AI), Data Processing Automation.

- **Technical & Engineering**: Geophysical & Geotechnical Investigation, MASW Inversion Modeling, Seismic Downhole Testing, Geoelectrical & Thermal Resistivity Survey.
- **Project & Operations**: Multidisciplinary Team Leadership (teams of 10+ personnel), HSE Implementation (Zero LTI), CAPEX/OPEX Cost & Resource Planning, Stakeholder Management.
- **Awards**: 1st place Kahf IA-ITB CUP 2025; Best XI player Kahf IA-ITB CUP 2025; 3rd place IA-ITB CUP 2024
- **Languages**: English (Fluent), Bahasa Indonesia (Native), Arabic (Fluent)